

Choosing Expanded or Factored Form

Answer Key

Algebra 1

Quick Answers

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|--|---|
| 1. $x^2 + 5x - 36$ | 8. $x(x - 2)(x + 2)$ |
| 2. $(x + 3)(x + 7)$ | 9. (a) constant term = 15, (b) = — |
| 3. 0, 4 s | 10. $(2x + 3)(3x - 4)$ |
| 4. $3x^2 - 15x + 2$ | 11. (a) simplified = $2x^2 + 6x$, (b) factored form = $2x(x + 3)$ |
| 5. $2(x - 3)(x + 3)$ | 12. (a) factored form = $(x + 2)(x + 5)$, (b) area at $x = 3 = 40 \text{ m}^2$, (c) = — |
| 6. (a) factored form = $(x - 2)(x - 4)$, (b) the zeros = 2, 4 | |
| 7. $4x^2 - 12x + 9$ | |
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What is verified

15 of 17 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-APR.A.1 – *problems 1, 4, 7, 9, 11, 12*

HSA-APR.B.3 – *problems 3, 6*

HSA-SSE.B.3 – *problems 2, 5, 6, 8, 10, 11, 12*

Difficulty 1–5 of 5 across 17 tagged checks.

- 1.** Write $(x - 4)(x + 9)$ in standard form.

Solution: Four products, then collect the middle pair: $x^2 + 9x - 4x - 36 = x^2 + 5x - 36$. The constant term -36 is the product of the two constants, and it is invisible until the brackets come off.

$x^2 + 5x - 36$

- 2.** Write $x^2 + 10x + 21$ in factored form.

Solution: Two numbers with product 21 and sum 10: since $21 = 3 \times 7$ and $3 + 7 = 10$, the factors are $(x + 3)$ and $(x + 7)$.

$(x + 3)(x + 7)$

- 3.** $h(t) = -5t(t - 4)$ metres: find both times at ground level.

Solution: The height is already factored, and a product is zero exactly when one of its factors is zero. So $-5t = 0$ gives $t = 0$, and $t - 4 = 0$ gives $t = 4$. No expanding and no quadratic formula is needed — that is the whole advantage of the factored form here. The ball leaves the ground at $t = 0$ and lands at $t = 4$.

$$t = 0 \text{ s} \quad \text{or} \quad t = 4 \text{ s}$$

4. Write $3x(x - 5) + 2$ in standard form.

Solution: Distribute first: $3x \cdot x = 3x^2$ and $3x \cdot (-5) = -15x$. The $+2$ is outside the product and simply follows: $3x^2 - 15x + 2$.

$$3x^2 - 15x + 2$$

5. Write $2x^2 - 18$ in fully factored form.

Solution: Common factor first: $2x^2 - 18 = 2(x^2 - 9)$. What is left is a difference of two squares, $x^2 - 3^2$, which factors as $(x - 3)(x + 3)$.

$$2(x - 3)(x + 3)$$

Skipping the common factor and writing $(2x - 6)(x + 3)$ is not fully factored; the first bracket still has a 2 in it.

6. $g(x) = x^2 - 6x + 8$: (a) rewrite to show the zeros; (b) give the zeros.

Solution (a): Two numbers with product 8 and sum -6 are -2 and -4 .

$$(x - 2)(x - 4)$$

Solution (b): A product is zero exactly when a factor is zero, so $x - 2 = 0$ or $x - 4 = 0$.

$$x = 2 \quad \text{or} \quad x = 4$$

7. Write $(2x - 3)^2$ in standard form.

Solution: A squared binomial is the bracket times itself, so it has three terms: $(2x)^2 = 4x^2$, twice the cross product $2 \cdot 2x \cdot (-3) = -12x$, and $(-3)^2 = 9$.

$$4x^2 - 12x + 9$$

8. Write $x^3 - 4x$ in fully factored form.

Solution: Every term has an x , so take it out first: $x^3 - 4x = x(x^2 - 4)$. The bracket is a difference of two squares: $x^2 - 2^2 = (x - 2)(x + 2)$.

$$x(x - 2)(x + 2)$$

9. $P(x) = (x + 5)(x - 1)(x - 3)$: (a) the constant term; (b) which form and why.

Solution (a): The constant term of a polynomial is its value at $x = 0$, because every other term carries at least one factor of x . Substituting into the factored form: $P(0) = (0 + 5)(0 - 1)(0 - 3) = (5)(-1)(-3)$.

$$15$$

Solution (b) — instructor-judged. The factored form did the work. Substituting $x = 0$ into three short brackets is three easy products; reading the constant term out of standard form would have meant expanding the whole cubic first. Full credit compares the effort the two forms demand; repeating the value 15 is not an explanation.

10. Write $6x^2 + x - 12$ in factored form.

Solution: With a leading coefficient of 6, split the middle term. Two numbers with product $6 \times (-12) = -72$ and sum 1 are 9 and -8 :

$$\begin{aligned} 6x^2 + x - 12 &= 6x^2 + 9x - 8x - 12 \\ &= 3x(2x + 3) - 4(2x + 3) = (2x + 3)(3x - 4). \end{aligned}$$

$(2x + 3)(3x - 4)$

11. $(x + 3)(x - 3) + (x + 3)^2$: (a) expand and simplify; (b) factor the result.

Solution (a): $(x + 3)(x - 3) = x^2 - 9$ and $(x + 3)^2 = x^2 + 6x + 9$. Adding them, the -9 and $+9$ cancel: $2x^2 + 6x$. $2x^2 + 6x$

Solution (b): Both terms share $2x$: $2x^2 + 6x = 2x(x + 3)$. Notice the shortcut the factored view offers — $(x + 3)$ was a factor of both starting pieces all along. $2x(x + 3)$

12. Garden of area $A(x) = x^2 + 7x + 10$ square metres: (a) factor; (b) area at $x = 3$; (c) which form gives the sides?

Solution (a): Two numbers with product 10 and sum 7 are 2 and 5.

$(x + 2)(x + 5)$

Solution (b): Substituting $x = 3$ into either form gives the same number. Using standard form: $9 + 21 + 10 = 40$. Using the factored form as a check: $(3 + 2)(3 + 5) = 5 \times 8 = 40$.

40 m^2

Solution (c) — instructor-judged. The factored form. A rectangle's area is the product of its two sides, and $(x + 2)(x + 5)$ is exactly that product written down, so the sides are $(x + 2)$ m and $(x + 5)$ m — at $x = 3$, 5 m by 8 m. Standard form gives no sides at all until it is factored. Full credit ties the two factors to the two side lengths; giving the numbers without that reason earns half credit.